

Drift-Regularized Multi-Token Prediction via On-Policy Self-Distillation

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Abstract

Standard autoregressive language models incur substantial latency by generating tokens strictly sequentially. Multi-token prediction (MTP) addresses this by predicting a block of future tokens in a single forward pass. However, training MTP models solely via token-level cross-entropy leaves a representation gap: the parallel hidden states lack explicit guidance on how to organize intermediate predictive representations, leading to chunk-level incompatibility and severe performance degradation over longer prediction horizons. While feature-regression techniques such as paired mean squared error (MSE) supply dense representation signals, they impose a rigid one-to-one Euclidean alignment that overconstrains the student hidden states. In this paper, we present a drift-regularized self-distillation framework for MTP. By constructing a local conditional attraction-repulsion field, our method transports student hidden state populations toward teacher feature distributions without forcing point-to-point Euclidean matching. We hypothesize and demonstrate that performance follows the strict ordering $\text{Perf}(\text{MTP}) < \text{Perf}(\text{MTP} + \text{MSE}) < \text{Perf}(\text{MTP} + \text{Drift})$ across benchmark accuracy, calibration, latent-space distribution alignment, and throughput-versus-latency efficiency.

1 Introduction

Standard next-token prediction (NTP) generates one token per step, incurring high memory bandwidth overhead and severe inference latency during long-form generation and reasoning-intensive tasks. Multi-token prediction (MTP) mitigates this bottleneck by reformulating decoding to generate a contiguous block of k tokens in a single forward pass [1]. In the on-policy self-distillation framework, a student model proposes future token candidates at inserted mask positions, while a frozen teacher evaluates the prefix to supply token-level supervision [1].

Despite its operational efficiency, optimizing MTP strictly via token-level cross-entropy leaves a critical representation gap. A correct token prediction constrains only the relative logit margins; infinitely many hidden state

configurations can yield identical argmax token outputs. Consequently, newly inserted MTP mask states must learn parallel future-token computations without explicit supervision over their representation geometry. Prior works on feature-based speculative drafting, such as EAGLE [2] and DFlash [9], explicitly highlight this instability, warning that representation uncertainty and cumulative feature errors heavily constrain long-horizon parallel generation. In MTP, this manifests as chunk-level incompatibility and rapid accuracy degradation as the prediction horizon k increases [1].

To address representation instability, direct feature matching via paired Mean Squared Error (MSE) serves as a baseline. However, paired MSE assumes a rigid one-to-one alignment under a fixed isotropic Gaussian assumption. This forces each parallel student state to replicate a single sequential teacher state in Euclidean space, which can overconstrain the parallel architecture when multiple feature neighborhoods yield valid output distributions, leading to conditional averaging and feature collapse.

To resolve the one-to-one matching limitation, we propose a conditional drifting regularizer for MTP. Based on population-level transport theory [4], our objective constructs a local vector field that attracts student queries toward target teacher feature neighborhoods while repelling them from overrepresented student clusters. This nonparametric alignment preserves local teacher modes and enforces coherent block-level feature geometry without forcing point-to-point Euclidean rigidity.

We evaluate the central hypothesis that representation regularization yields consistent performance gains:

$$\text{Perf}(\text{MTP}) < \text{Perf}(\text{MTP} + \text{MSE}) < \text{Perf}(\text{MTP} + \text{Drift}) \quad (1)$$

Crucially, outperformance is assessed across a comprehensive set of metrics:

1. **Task Accuracy:** Evaluated on reasoning benchmarks under static block sizes k .
2. **Calibration:** Measured via Expected Calibration Error (ECE) to assess whether token-level confidence tracks empirical correctness.

Progression of Multi-Token Prediction Formulations

Objective	Core Representation Mechanism & Inductive Bias
$\mathcal{L}_{\text{MTP}}$ (Native)	Supervises output logits only. Suffers from representation ambiguity and error accumulation over horizon k .
$\mathcal{L}_{\text{MTP}+\text{MSE}}$	Pointwise regression to teacher states. Imposes strict one-to-one Euclidean match; risks overconstraint.
$\mathcal{L}_{\text{MTP}+\text{Drift}}$ (Proposed)	Population-level attraction-repulsion transport. Preserves local modes and enhances chunk-level compatibility.

Figure 1: Summary of MTP loss formulations evaluated in this study.

3. **Inference Efficiency:** Characterized through Throughput vs. Latency Pareto curves under varying static k configurations.

2 Preliminary & Motivation

2.1 Flattened On-Policy MTP Self-Distillation

Let a training sequence region r contain a context prefix c_r and request k_r future token predictions. The flattened student input inserts $k_r - 1$ trainable <MTP> mask tokens. Repeated position IDs and a blocked causal mask enable each active state to attend to the prefix and preceding masks without altering copied context representations [1].

Let $\mathcal{P}_{r,j}$ denote the flattened position predicting horizon $j \in \{1, \dots, k_r\}$. The student state, logit vector, and rollout token are:

$$u_{r,j}^S = H_\theta(\tilde{e})[\mathcal{P}_{r,j}], \quad (2)$$

$$Z_{r,j}^S = W u_{r,j}^S + b, \quad (3)$$

$$\hat{y}_{r,j} = \arg \max_{v \in \mathcal{V}} Z_{r,j,v}^S. \quad (4)$$

The frozen teacher model is evaluated under the student-forced continuation:

$$u_{r,j}^T = H_{\theta_T}(c_r, \hat{y}_{r,<j}), \quad (5)$$

$$\bar{y}_{r,j} = \arg \max_{v \in \mathcal{V}} p_{\theta_T}(v \mid c_r, \hat{y}_{r,<j}). \quad (6)$$

The hard on-policy distillation cross-entropy loss is defined over active MTP positions $N_{\text{act}} = \sum_r k_r$:

$$\mathcal{L}_{\text{MTP}} = -\frac{1}{N_{\text{act}}} \sum_{r,j} \log p_{r,j}^S(\bar{y}_{r,j}). \quad (7)$$

2.2 Unified Latent Notation

Let ϕ_S and ϕ_T be feature mappings into a shared d -dimensional space (e.g., post-layer-normalization hidden states). For an active position $i = (r, j)$:

$$x_i = \phi_S(u_i^S), \quad y_i^+ = \phi_T(u_i^T), \quad y_i^- = \text{sg}(\phi_S(u_i^S)), \quad (8)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. The current student population serves as negative samples y^- after detachment and self-exclusion.

2.3 Motivation for Representation Regularization

While Eq. (7) optimizes final logit distributions, it leaves the internal feature representations $u_{r,j}^S$ unconstrained. In multi-token decoding, an unregularized student often produces tokens that are individually plausible according to marginal probability distributions but jointly incompatible across the block k . As k increases, compounding feature errors degrade output accuracy. This motivates introducing direct feature-level supervision to enforce coherence, improve token calibration, and extend the usable static block size k during deployment.

3 Background & Theoretical Justification

3.1 Why Output-Only Parallel Prediction Needs Latent Regularization

For a target token y and an unembedding matrix W with row vectors w_v , any hidden state h residing in the convex cone:

$$\mathcal{C}_y(m) = \{h \in \mathbb{R}^d : (w_y - w_v)^\top h \geq m, \forall v \neq y\} \quad (9)$$

achieves a logit margin of at least m . Token cross-entropy supervision identifies a broad feasible subspace $\mathcal{C}_y(m)$ without organizing the representation geometry. When newly inserted mask states must approximate a sequential computation, output-only loss fails to constrain intermediate trajectory dynamics. Feature-based speculative drafting literature (e.g., EAGLE [2], NextLat [3]) corroborates that explicit latent regression is critical to stabilize long-horizon future representations.

3.2 Pointwise Latent Regression: Paired MSE

The simplest auxiliary regularizer is pointwise paired MSE:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N_{\text{act}}} \sum_i \|x_i - \text{sg}(y_i^+)\|_2^2. \quad (10)$$

Probabilistically, assuming isotropic Gaussian conditional distributions $q^T(z \mid c_i) = \mathcal{N}(y_i^+, \sigma^2 I)$ and $q^S(z \mid c_i) =$

Table 1: Comparison of latent regularizers.

Property	Paired MSE	Conditional Drift
Matching Unit	1-to-1 Target Pair	Local Population
Probabilistic View	Gaussian KL	Nonparametric Transport
Primary Bias	Euclidean Alignment	Kernel Affinity
Primary Risk	Overconstraint / Collapse	Noisy Mini-batch Pool
Compute Cost	$\mathcal{O}(N)$	$\mathcal{O}(N \cdot I^+)$

$\mathcal{N}(x_i, \sigma^2 I)$, Eq. (10) equals the conditional Gaussian KL divergence:

$$D_{\text{KL}}(q^T \parallel q^S) = \frac{1}{2\sigma^2} \|y_i^+ - x_i\|_2^2. \quad (11)$$

Although $\mathcal{L}_{\text{MSE}}$ provides dense supervision, its underlying assumption—that every student state x_i must map strictly to a single teacher state y_i^+ under Euclidean distance—is restrictive. When parallel mask representations naturally deviate from sequential teacher states while maintaining distributional equivalence, paired MSE induces overconstraint and feature averaging.

3.3 Drifting as Population-Level Regularization

The drifting framework [4] replaces pointwise matching with population-level transport. It constructs a vector field $V_{p,q}(x)$ between a target teacher population p and a model student population q_θ :

$$V_{p,q}(x) = \frac{\mathbb{E}_{y^+ \sim p}[\kappa(x, y^+)(y^+ - x)]}{\mathbb{E}_{y^+ \sim p}[\kappa(x, y^+)]} - \frac{\mathbb{E}_{y^- \sim q}[\kappa(x, y^-)(y^- - x)]}{\mathbb{E}_{y^- \sim q}[\kappa(x, y^-)]}, \quad (12)$$

where $\kappa(\cdot, \cdot)$ is a localized kernel. The objective regresses student state x toward a stopped transported target:

$$\mathcal{L}_{\text{drift}} = \mathbb{E}_{x \sim q_\theta} [\|x - \text{sg}(x + \eta V_{p,q_\theta}(x))\|_2^2]. \quad (13)$$

By pulling queries toward nearby teacher clusters (attraction) and pushing them away from crowded student regions (repulsion), drifting preserves multi-modal feature distributions and prevents state collapse.

4 Methodology

4.1 Unified Regularized Objective

We formulate the unified regularized MTP objective as:

$$\mathcal{L} = \mathcal{L}_{\text{MTP}} + \lambda_R \mathcal{L}_R, \quad R \in \{\text{MSE}, \text{drift}\}. \quad (14)$$

4.2 Conditional Drift Estimator

For each active MTP query index $i = (r, j)$, we construct a positive teacher pool $\mathcal{I}_i^+$ and a negative student pool $\mathcal{I}_i^-$. We define a conditional kernel:

$$\kappa_{\text{cond}}(i, l; y_l) = \kappa_h(x_i, y_l) \cdot \kappa_c(g(c_i), g(c_l)) \cdot \mathbf{1}[a_i = a_l], \quad (15)$$

where $a_i = j$ restricts interaction to identical prediction horizons, $g(c)$ represents detached prefix context embeddings, and κ_h is a normalized distance kernel.

Normalized affinity weights are computed as:

$$a_{il}^+ = \frac{\kappa_{\text{cond}}(i, l; y_l^+)}{\sum_{m \in \mathcal{I}_i^+} \kappa_{\text{cond}}(i, m; y_m^+) + \epsilon}, \quad (16)$$

$$a_{il}^- = \frac{\kappa_{\text{cond}}(i, l; y_l^-)}{\sum_{m \in \mathcal{I}_i^- \setminus \{i\}} \kappa_{\text{cond}}(i, m; y_m^-) + \epsilon}. \quad (17)$$

Removing query self-interaction, the estimated drift vector is:

$$\hat{V}_i = \sum_{l \in \mathcal{I}_i^+} a_{il}^+ (y_l^+ - x_i) - \sum_{l \in \mathcal{I}_i^- \setminus \{i\}} a_{il}^- (y_l^- - x_i). \quad (18)$$

The auxiliary drift loss is implemented via a stopped-target update:

$$\mathcal{L}_{\text{drift}}^{\text{cond}} = \frac{1}{N_{\text{act}}} \sum_i \left\| x_i - \text{sg} \left(x_i + \eta \frac{\hat{V}_i}{\|\hat{V}_i\|_2 + \epsilon} \right) \right\|_2^2. \quad (19)$$

4.3 Horizon-Wise and Block-Level Variants

In horizon-wise drift ($\mathcal{L}_{\text{drift, horizon}}$), alignment groups enforce $a_i = j$. To enforce joint chunk coherence across the entire block, we also evaluate block-level drift ($\mathcal{L}_{\text{drift, block}}$) using concatenated normalized block representations:

$$b_r^T = \frac{1}{\sqrt{k_r}} \text{vec}[x_{r,1}; \dots; x_{r,k_r}], \quad b_r^T = \frac{1}{\sqrt{k_r}} \text{vec}[y_{r,1}^+; \dots; y_{r,k_r}^+]. \quad (20)$$

4.4 Gradient and Compute Matching

To evaluate $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_{\text{drift}}$ fairly, we track the gradient magnitude ratio ρ_R across shared model parameters θ :

$$\rho_R = \frac{\mathbb{E} \|\lambda_R \nabla_\theta \mathcal{L}_R\|_2}{\mathbb{E} \|\nabla_\theta \mathcal{L}_{\text{MTP}}\|_2}, \quad R \in \{\text{MSE}, \text{drift}\}. \quad (21)$$

Loss coefficients λ_{MSE} and λ_{drift} are adjusted during warm-up to match ρ_R , ensuring that gains stem from inductive bias rather than differential gradient scaling.

5 Experiments & Results Setup

5.1 Experimental Setup and Protocols

Training is performed on mathematical reasoning traces from GSM8K / MetaMathQA datasets. Models are trained under static block sizes $k \in \{2, 3, 4, 5\}$.

Evaluation is conducted across a suite of reasoning and generation benchmarks: GSM8K (grade-school math), AIME25 (competition math), GPQA (graduate science), BBH (Big Bench Hard), IFEval (instruction following), and CNN/DailyMail (long-form summarization).

5.2 Hyperparameter Tuning Strategy

To isolate the effect of loss weighting λ and static block size k , we follow a two-stage evaluation protocol:

1. **Stage 1 (λ Sweep at Fixed $k = 3$):** Fix $k = 3$ and sweep loss coefficients $\lambda \in [10^{-4}, 10^{-1}]$ for both MSE and Drift objectives to determine the optimal coefficient matching $\rho_R \approx 0.10$.
2. **Stage 2 (Static k Sweep):** Fix λ at its optimal value and vary static chunk size $k \in \{2, 3, 4, 5\}$ to establish accuracy and latency trade-off curves.

5.3 Evaluation Metrics

Models are evaluated across four complementary dimensions.

Task performance and forgetting. We report exact-match accuracy for GSM8K, AIME25, GPQA Main, and BBH, prompt-level exact-match for IFEval, and ROUGE-L for CNN/DailyMail. To make capability retention explicit, the original frozen teacher is evaluated with ordinary sequential ($k = 1$) decoding and is shown as a horizontal reference in accuracy-versus-acceleration plots. The vertical gap between an MTP point and this teacher reference is the empirical “forgetting” at that operating point. This is a more direct test of the central claim than comparing MTP variants only with one another.

Expected Calibration Error (ECE). Following the token-level calibration analysis of Joshi et al. [5], ECE measures whether predicted confidence agrees with empirical correctness. Given confidence bins B_m , we use

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (22)$$

The referenced study applies ECE to token predictions obtained by projecting intermediate residual representations through the unembedding matrix, and uses ECE together with accuracy to expose calibration changes that can occur even when accuracy has already saturated [5]. We follow the same token-level interpretation at the final prediction head. Lower ECE is better. Importantly, ECE is a calibration metric rather than a direct student–teacher distance; the latent-space tests below are therefore used as the explicit distribution-matching diagnostics.

Latent-space distribution alignment. On a held-out validation set, we compare pooled student and teacher feature distributions at matched prediction horizons. Features are compared directly in their common latent space, **with the same normalization applied to both populations when required for distance-based statistics?** We report four complementary statistics, all lower-is-better except 1-NN accuracy, for which 50% is the ideal indistinguishable limit:

1. **Sliced Wasserstein-2 (SW_2):** the average one-dimensional Wasserstein-2 distance over a set of random projections.

2. **Maximum Mean Discrepancy (MMD):** a kernel-based two-sample discrepancy between the student and teacher feature populations.
3. **Sinkhorn distance:** an entropically regularized optimal-transport discrepancy between the two empirical feature distributions.
4. **1-NN two-sample accuracy (NN1):** a classifier-free distinguishability test formed by pooling student and teacher samples and assigning each sample the label of its nearest neighbor. Values close to 50% indicate that the two populations are difficult to distinguish.

These statistics provide complementary views of distributional agreement: SW_2 measures geometric discrepancy through random projections, MMD captures kernel-based distributional differences, Sinkhorn measures the transport cost required to align the two populations, and NN1 directly quantifies their distinguishability. The choice of these diagnostics follows the use of complementary distributional measures in Kernel-Gradient Drifting Models [6]. For our MTP setting, all quantities are computed separately at each prediction horizon j and summarized over $j = 1, \dots, 7$. We retain the horizon-wise curves on the validation set to test whether distributional mismatch is reduced at later prediction horizons, where representation drift and forgetting are expected to accumulate.

Inference efficiency. We report realized acceleration k_{eff} and deployment throughput versus end-to-end latency. The training-time regularizers add no inference module, so system-level differences should arise primarily from accepted block length and adaptive-control behavior.

5.4 Evaluated Model Variants

We evaluate six MTP objective variants:

- **M0 (Hard On-Policy MTP):** Native token-only distillation cross-entropy loss (Baseline).
- **M1 (M0 + Paired Latent MSE):** Adds pointwise hidden state regression.
- **M2 (M0 + Conditional Full Drift):** Primary proposed objective combining attraction and repulsion fields.
- **M3 (Removed?: M0 + Attraction-Only Drift):** Ablation disabling the negative repulsion field ($\mathcal{I}^-$).
- **M4 (M0 + Block-Level Full Drift):** Applies drift transport on concatenated block representations b_r .
- **M5 (Soft On-Policy KL):** Replaces hard token target $\bar{y}$ with full teacher distribution KL divergence.

Table 2: MTP experimental variants.

ID	Objective Formulation	Diagnostic Role
M0	$\mathcal{L}_{\text{MTP}}$	Token-only Baseline
M1	$\mathcal{L}_{\text{MTP}} + \lambda_M \mathcal{L}_{\text{MSE}}$	Pointwise Latent Signal
M2	$\mathcal{L}_{\text{MTP}} + \lambda_D \mathcal{L}_{\text{drift, horiz}}^{\text{cond}}$	Full Conditional Transport
M3	$\mathcal{L}_{\text{MTP}} + \lambda_D \mathcal{L}_{\text{attract}}$	Repulsion Ablation
M4	$\mathcal{L}_{\text{MTP}} + \lambda_D \mathcal{L}_{\text{drift, block}}^{\text{cond}}$	Joint Block Coherence
M5	$\mathcal{L}_{\text{MTP, soft-KL}}$	Distributional Token Baseline

5.5 Empirical Hypothesis & Decision Rules

We test the explicit performance ordering:

$$\text{Perf}(M0) < \text{Perf}(M1) < \text{Perf}(M2).$$

The decision rules and expected qualitative diagnostic trends are summarized in Table 3.

Table 3: Experimental decision rules and interpretation.

Observed Outcome	Mechanistic Interpretation
$M0 \geq M1, M2$	Token loss sufficient; latent regularizers add excess bias or noise.
$M1 > M0, M2$	Pointwise alignment is optimal; population fields introduce sampling noise.
$M2 > M1 > M0$	Population transport resolves 1-to-1 alignment defects and prevents state collapse.
$M2 \approx M3$	Attraction drives gains; repulsion signal is redundant.
$M4 > M2$	Inter-token feature dependency is a dominant bottleneck.

5.6 Revision-2 Evaluation Matrix and Diagnostic Plots

Draft status. All numerical values in this revision are synthetic placeholders used to establish the reporting structure and the intended qualitative story. They must be replaced by measured results before submission. The main change from Revision 1 is that The previous average-confidence evaluation is removed entirely. Confidence is retained only as the control variable of the confidence-adaptive decoder. We instead separate three distinct questions: (i) how much capability is retained relative to the teacher, (ii) whether token confidence is calibrated against correctness, and (iii) whether the student and teacher latent populations actually become more similar.

Decoding sweep and teacher reference. We evaluate static decoding at $k \in \{1, \dots, 7\}$

and confidence-adaptive decoding over $\tau \in \{0.995, 0.99, 0.98, 0.97, 0.96, 0.95, 0.90, 0.87, 0.85, 0.80, 0.75, 0.70, 0.65, 0.60\}$. The realized acceleration is

$$k_{\text{eff}} = \frac{\text{emitted non-padding tokens}}{\text{decoding forward passes}}. \quad (23)$$

The frozen NTP teacher is evaluated under ordinary sequential decoding and its score is plotted as a horizontal $k=1$ reference. Hence, in accuracy-versus-acceleration plots, the vertical distance from an MTP curve to the teacher line has a direct interpretation as capability forgetting at that speed. The teacher line is a reference, not a trainable upper bound: the MTP model is allowed to trade a small amount of score for a much larger acceleration, but the goal of the regularizer is to shrink that gap.

Why ECE and latent distances are both needed.

ECE is intentionally not presented as a student-teacher distance. Joshi et al. [5] use ECE and MCE to study how confidence aligns with correctness, including cases where accuracy is already stable. In our setting, ECE therefore answers whether the student has reliable confidence at the chosen operating point. The latent statistics answer a different question: are the student hidden-state populations actually becoming indistinguishable from the teacher populations? Combining the two prevents us from incorrectly interpreting a low ECE as proof of latent distribution matching.

Accuracy versus acceleration with explicit forgetting reference. Figure 2 is the primary result. Static points use square markers; adaptive settings use circular markers. The black dashed line is the frozen teacher score. A stronger regularizer should shift the MTP frontier upward toward this line at the same k_{eff} .

Static block-size diagnostics. Figure 3 retains the configured block size on the horizontal axis. The accuracy panel again includes the teacher reference, while the second panel shows the actual acceleration and the third panel shows ECE. This figure separates configured compute expansion from the resulting quality and calibration cost.

Confidence-adaptive decoding: score retention and calibration. Figure 4 removes the previous confidence panel. Instead, it shows accuracy retention to the teacher and ECE at the same measured acceleration. This directly tests whether a regularizer can move the Pareto frontier upward without merely producing a more confident but equally forgetful model.

Throughput-latency frontier. We additionally measure deployment efficiency under the same serving stack and hardware. Figure 5 uses end-to-end latency in milliseconds per generated token on the horizontal axis and

Table 4: Decoding settings and quantities reported in Revision 2. The previous average-confidence evaluation is removed. Token confidence is used internally only by the confidence-adaptive decoder.

Decoding mode	Evaluated settings	Reported quantities
Static	$k \in \{1, 2, 3, 4, 5, 6, 7\}$	score, k_{eff} , ECE
Confidence-adaptive	full τ grid above	score, k_{eff} , ECE
Validation latent diagnostics	$j \in \{1, \dots, 7\}$	SW ² , MMD, Sinkhorn, NN1
Deployment	$c \in \{1, 2, 4, 8, 16\}$	throughput, latency

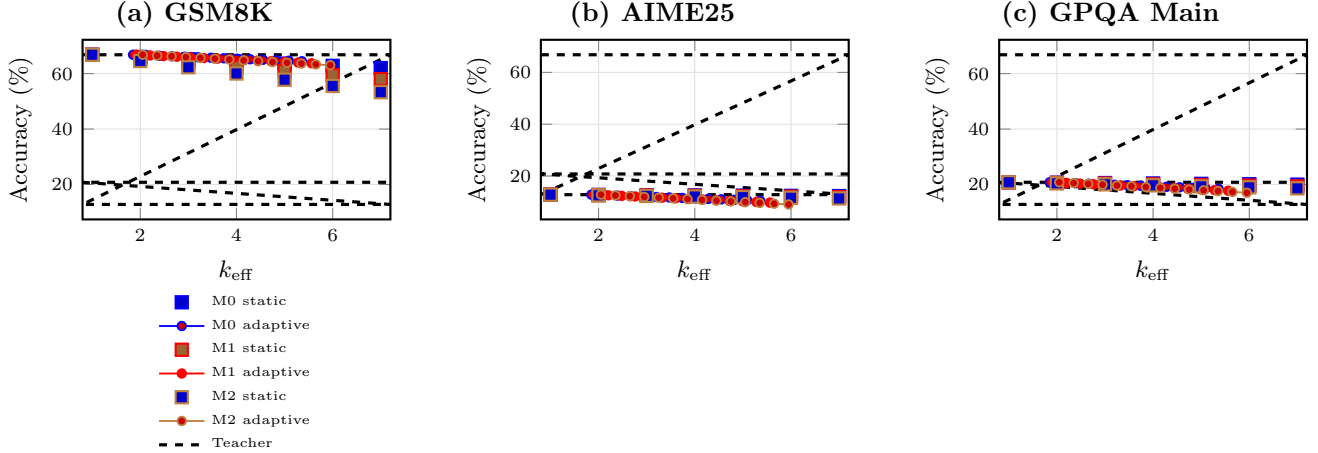


Figure 2: Accuracy versus measured acceleration. The dashed teacher reference makes forgetting visually explicit: at any common k_{eff} , smaller vertical separation from the teacher indicates better retention. Values are synthetic placeholders.

aggregate throughput on the vertical axis. Lower latency and higher throughput are better.

Horizon-wise degradation and teacher reference.

Figure 6 directly tests whether latent regularization reduces compounding forgetting at later prediction positions. The teacher curve is the sequential NTP token-accuracy reference evaluated on the same held-out token positions; it is plotted as a horizontal reference because the teacher does not have an MTP horizon. ECE is shown separately because it measures calibration, not teacher-student feature distance.

Latent-space distribution alignment on validation data.

The central mechanistic diagnostic is performed directly on hidden-state populations. For each held-out MTP query, the student feature x_i and teacher feature y_i^+ are normalized in a shared space; interactions and validation statistics are restricted to matched prediction horizons. We then compute SW², MMD, Sinkhorn, and NN1 between the student and teacher populations. The exact statistics are chosen to be complementary: SW² is sensitive to projected transport, MMD measures kernel mean-embedding discrepancy, Sinkhorn provides an entropically regularized transport cost, and NN1 measures how easily a classifier can distinguish the two pooled populations [6]. A successful method should move all three

distance-like metrics downward and NN1 toward 0.5.

Cross-benchmark operating-point summary. To make the teacher-retention story visible beyond the three primary plots, the final test summary reports the same score/ k_{eff} /ECE triplet across all six benchmark categories. The table uses a single illustrative adaptive operating point ($\tau = 0.90$ in the synthetic data); the actual paper should use the validation-selected operating point without tuning on test data.

Latent-space final table. Table 6 collects the final held-out validation values for the four distribution diagnostics. A useful experimental outcome is not simply that M2 has the best benchmark score; the student population should also be measurably harder to distinguish from the teacher population than M0.

Choosing λ_R by gradient matching. The regularization coefficient λ_R remains a training hyperparameter. During a short warm-up period at fixed $k = 3$, we maintain

$$\bar{r}_R = \text{EMA} \left[\frac{\|\nabla_{\theta} \mathcal{L}_R\|_2}{\|\nabla_{\theta} \mathcal{L}_{\text{MTP}}\|_2 + \epsilon} \right], \quad \lambda_R = \frac{\rho^*}{\bar{r}_R}. \quad (24)$$

After warm-up, λ_R is frozen. We continue logging ρ_R to verify that MSE and drift are compared at approximately matched auxiliary gradient strength.

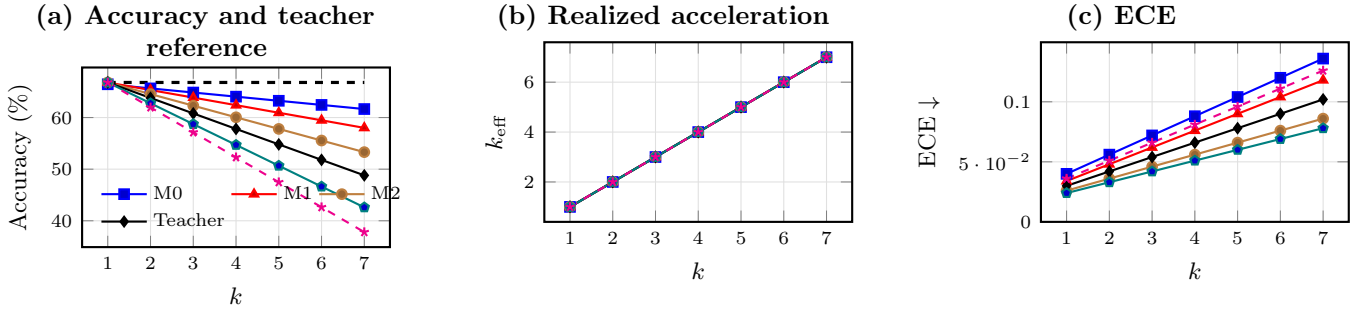


Figure 3: Static- k diagnostics on GSM8K for all objective variants. The teacher reference is shown only in the accuracy panel; it has no configured MTP block size. Values are synthetic placeholders.

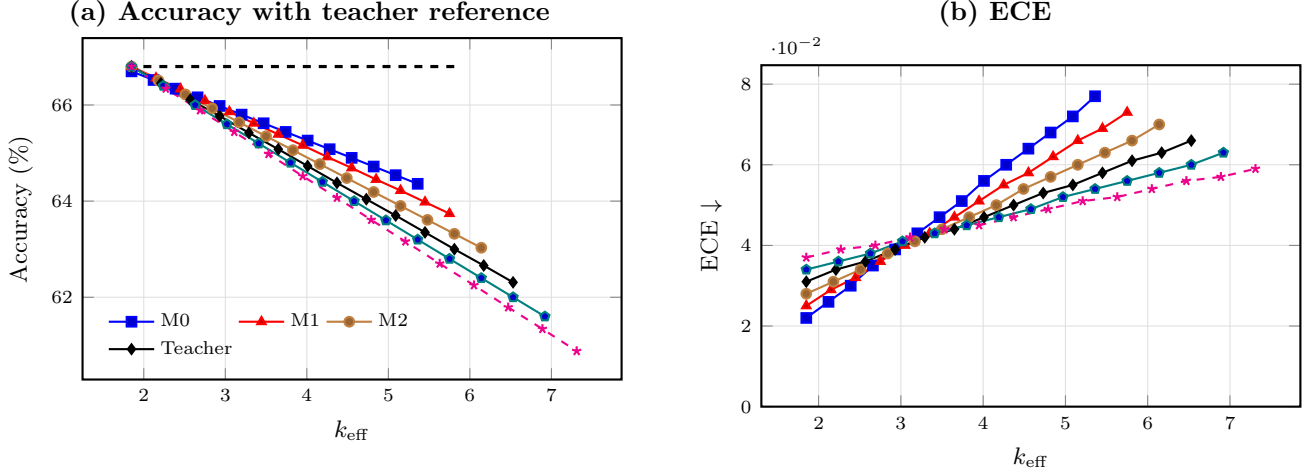


Figure 4: Confidence-adaptive decoding on GSM8K. The teacher reference provides a common zero-forgetting target, while ECE reveals whether the student confidence is calibrated at the same operating point. Values are synthetic placeholders.

Selection and reporting rule. The primary result remains the complete accuracy–acceleration curve. For a single-number summary, the operating point is selected on validation as the fastest setting whose validation score is no more than one percentage point below sequential teacher scoring. The same setting is then frozen and evaluated once on the test set. Accuracy/score, k_{eff} , and ECE are reported together. Latent-space distribution metrics are measured on a held-out validation split and are not tuned on test data. This separation keeps capability retention, calibration, and feature-space alignment as distinct diagnostics.

6 Related Work & Future Work

6.1 Related Work

Multi-token prediction and speculative drafting aim to accelerate inference by breaking the single-token bottleneck. MTP models predict multiple future tokens directly through parallel architecture heads [1]. Speculative decoding systems, such as EAGLE [2] and Medusa, use auxiliary drafting heads trained alongside a target model.

EAGLE uses Smooth L1 feature regression to guide draft predictions, explicitly pointing out feature uncertainty as a primary performance bound.

6.2 Future Work: Application to Speculative Drafting (DFlash)

While this paper focuses on standalone MTP, the conditional drifting regularizer is applicable to speculative drafting frameworks such as DFlash [9].

DFlash uses a block-diffusion drafter q_θ trained with position-weighted token cross-entropy:

$$\mathcal{L}_{\text{DF}} = -\frac{1}{N_{\text{mask}}} \sum_{r,j \in \mathcal{M}_r} \omega_j \log q_\theta(y_{r,j} | \tilde{e}_r, g_T(c_r)), \quad (25)$$

where ω_j decays with draft depth j .

We propose extending DFlash with conditional drifting:

$$\mathcal{L}_{\text{DFlash+Drift}} = \mathcal{L}_{\text{DF}} + \lambda_D \mathcal{L}_{\text{drift, DF}}^{\text{cond}}, \quad (26)$$

where draft state $x_{r,j,\nu}^{\text{DF}}$ is aligned with target feature $y_{r,j}^{+, \text{DF}}$ under alignment group $a_i = (j, \nu)$ representing draft position j and corruption step ν .

Table 5: Synthetic cross-benchmark summary at one common confidence-adaptive operating point. Each row reports task score, realized acceleration, and ECE; the teacher score is included as the retention reference.

Dataset	Variant	Score type	Score	k_{eff}	ECE	Teacher score
GSM8K	M0	Accuracy	58.8	3.3	0.06	66.8
GSM8K	M1	Accuracy	59.6	3.39	0.052	66.8
GSM8K	M2	Accuracy	60.5	3.57	0.045	66.8
GSM8K	M3	Accuracy	59.9	3.5	0.048	66.8
GSM8K	M4	Accuracy	60.8	3.63	0.043	66.8
GSM8K	M5	Accuracy	58.7	3.35	0.057	66.8
AIME25	M0	Accuracy	8.8	4.7	0.07	12.8
AIME25	M1	Accuracy	9.6	4.79	0.059	12.8
AIME25	M2	Accuracy	10.7	4.97	0.05	12.8
AIME25	M3	Accuracy	10.1	4.9	0.054	12.8
AIME25	M4	Accuracy	10.9	5.03	0.047	12.8
AIME25	M5	Accuracy	9.2	4.75	0.065	12.8
GPQA Main	M0	Accuracy	18.4	2.0	0.058	20.8
GPQA Main	M1	Accuracy	18.8	2.09	0.052	20.8
GPQA Main	M2	Accuracy	19.0	2.27	0.046	20.8
GPQA Main	M3	Accuracy	18.8	2.2	0.049	20.8
GPQA Main	M4	Accuracy	19.0	2.33	0.044	20.8
GPQA Main	M5	Accuracy	18.6	2.05	0.056	20.8
BBH	M0	Accuracy	56.0	2.7	0.067	60.5
BBH	M1	Accuracy	57.2	2.78	0.059	60.5
BBH	M2	Accuracy	58.5	2.95	0.051	60.5
BBH	M3	Accuracy	58.0	2.88	0.054	60.5
BBH	M4	Accuracy	58.8	3.0	0.05	60.5
BBH	M5	Accuracy	56.6	2.73	0.064	60.5
IFEval	M0	Accuracy	68.1	2.3	0.061	72.0
IFEval	M1	Accuracy	69.0	2.4	0.054	72.0
IFEval	M2	Accuracy	70.2	2.56	0.046	72.0
IFEval	M3	Accuracy	69.7	2.49	0.049	72.0
IFEval	M4	Accuracy	70.5	2.62	0.044	72.0
IFEval	M5	Accuracy	68.7	2.35	0.059	72.0
CNN/DailyMail	M0	ROUGE-L	33.2	1.7	0.081	35.0
CNN/DailyMail	M1	ROUGE-L	33.8	1.77	0.072	35.0
CNN/DailyMail	M2	ROUGE-L	34.3	1.92	0.064	35.0
CNN/DailyMail	M3	ROUGE-L	34.0	1.86	0.067	35.0
CNN/DailyMail	M4	ROUGE-L	34.4	1.96	0.061	35.0
CNN/DailyMail	M5	ROUGE-L	33.5	1.73	0.078	35.0

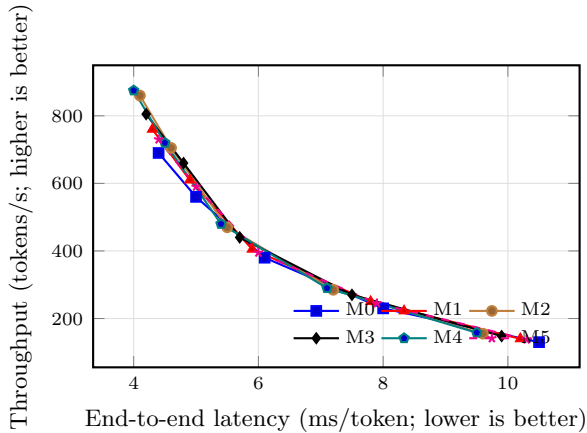


Figure 5: Synthetic throughput-latency curves at concurrency $c \in \{1, 2, 4, 8, 16\}$. The final version should state hardware, engine version, batch limits, warm-up policy, and control-flow execution mode.

In future work, we plan to evaluate whether $\mathcal{L}_{\text{DFlash+Drift}}$ increases average accepted draft length α and end-to-end token throughput compared to native DFlash ($\mathcal{L}_{\text{DF}}$) and paired MSE variants ($\mathcal{L}_{\text{DF}} + \lambda_M \mathcal{L}_{\text{MSE}}$).

Table 6: Synthetic final latent-space distribution diagnostics on the held-out validation set. Lower is better for SW^2 , MMD, and Sinkhorn; NN1 is best at 50%.

Variant	SW^2	MMD	Sinkhorn	NN1
M0	0.7	0.063	0.54	0.61
M1	0.51	0.046	0.42	0.58
M2	0.33	0.029	0.32	0.53
M3	0.43	0.038	0.38	0.55
M4	0.32	0.027	0.3	0.51
M5	0.67	0.057	0.52	0.6

7 Conclusion

This work addresses the representation gap in output-only multi-token prediction. We demonstrate that while paired MSE latent regression offers dense feature guidance, its rigid one-to-one Euclidean matching assumption can over-constrain parallel state representations. By introducing a conditional drift regularizer based on local attraction-repulsion transport, we align student feature distributions with teacher targets while preserving local feature manifolds. The proposed framework targets simultaneous improvements in benchmark accuracy retention, calibration, latent distribution alignment, and throughput-versus-latency efficiency under static and confidence-adaptive

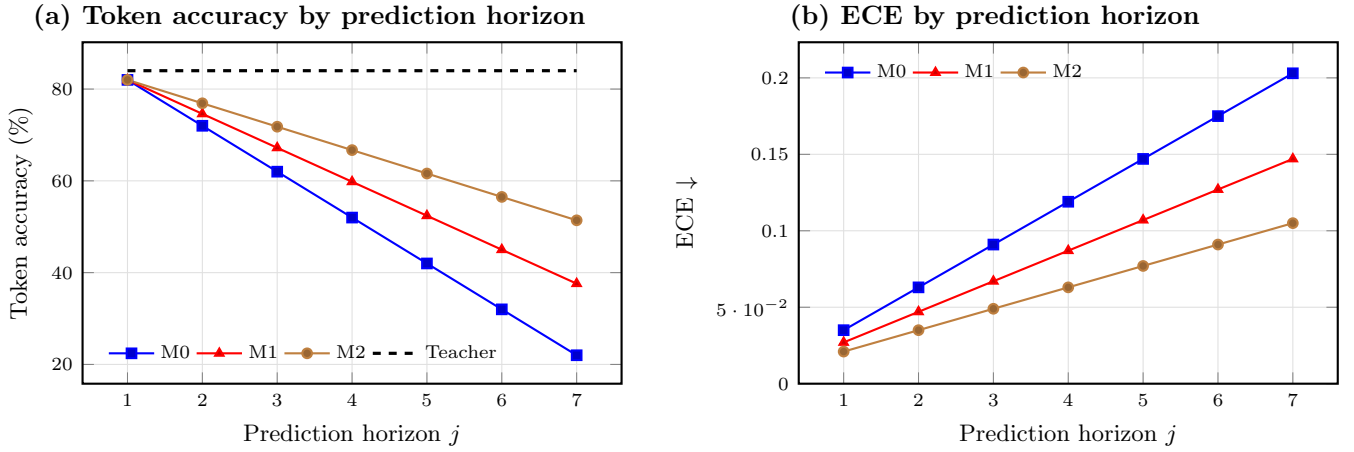


Figure 6: Horizon-wise degradation. The teacher reference makes late-horizon forgetting explicit; a successful regularizer should flatten the gap to the teacher rather than only improving the first prediction position. Values are synthetic placeholders.

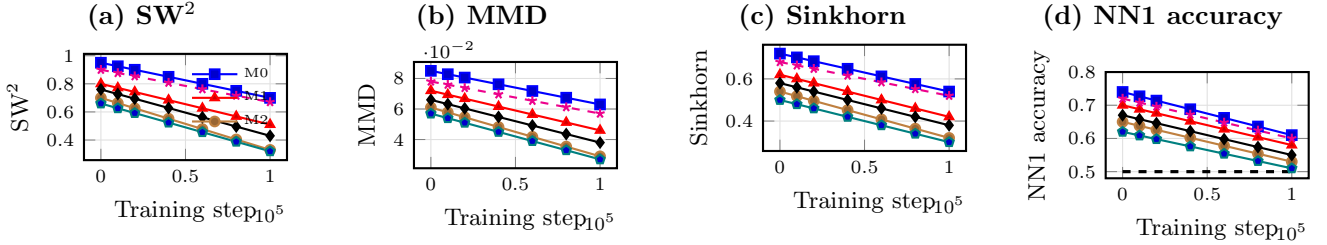


Figure 7: Synthetic validation curves for student-teacher latent-space alignment. Distance-like metrics decrease as distributions align, while NN1 approaches the 50% indistinguishability limit. The figure is meant to be replaced by held-out validation measurements; it is not a claim that the current placeholder values have been observed experimentally.

decoding.

Appendix: Complete Operating-Point Tables

Reporting note. The four latent metrics should be logged during training on the validation split, with per-horizon values available in the raw experiment logs. Final figures should report uncertainty over validation samples and, where feasible, over training seeds.

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Table 7: Complete static and confidence-adaptive sweep for Hard on-policy MTP (M0). Accuracy/score, k_{eff} , and ECE are reported together for the three primary benchmarks. All numerical values are synthetic placeholders.

Decoding setting	GSM8K			AIME25			GPQA Main		
	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE
Static k=1	66.0	1	0.032	12.0	1	0.049	20.0	1	0.046
Static k=2	60.7	2	0.046	11.2	2	0.059	19.0	2	0.055
Static k=3	53.0	3	0.06	10.2	3	0.069	17.0	3	0.064
Static k=4	49.8	4	0.074	9.2	4	0.079	15.0	4	0.073
Static k=5	41.5	5	0.088	8.0	5	0.089	13.0	5	0.082
Static k=6	36.5	6	0.102	6.8	6	0.099	11.0	6	0.091
Static k=7	32.0	7	0.116	5.6	7	0.109	9.0	7	0.1
Adaptive tau=0.995	64.5	1.9	0.04	12.7	2.85	0.055	20.6	1.5	0.051
Adaptive tau=0.99	63.9	2.1	0.041	12.3	3.05	0.057	20.3	1.5	0.052
Adaptive tau=0.98	63.2	2.3	0.043	11.8	3.25	0.058	19.9	1.6	0.053
Adaptive tau=0.97	62.4	2.5	0.044	11.4	3.45	0.06	19.6	1.8	0.055
Adaptive tau=0.96	62.0	2.6	0.046	11.0	3.55	0.062	19.2	1.9	0.056
Adaptive tau=0.95	61.1	2.8	0.047	10.6	3.75	0.064	18.9	2.1	0.058
Adaptive tau=0.90	58.6	3.3	0.048	10.2	4.25	0.065	18.5	2.6	0.059
Adaptive tau=0.87	56.9	3.6	0.05	9.7	4.55	0.067	18.2	2.9	0.06
Adaptive tau=0.85	56.3	3.7	0.051	9.3	4.65	0.069	17.8	3.0	0.062
Adaptive tau=0.80	54.3	4.0	0.053	8.9	4.95	0.07	17.5	3.3	0.063
Adaptive tau=0.75	52.3	4.3	0.054	8.5	5.25	0.072	17.1	3.6	0.065
Adaptive tau=0.70	50.1	4.6	0.055	8.1	5.55	0.074	16.8	3.9	0.066
Adaptive tau=0.65	47.0	5.0	0.057	7.6	5.95	0.075	16.4	4.3	0.067
Adaptive tau=0.60	44.5	5.3	0.058	7.2	6.25	0.077	16.1	4.6	0.069

Table 8: Complete static and confidence-adaptive sweep for MTP with paired latent MSE (M1). Accuracy/score, k_{eff} , and ECE are reported together for the three primary benchmarks. All numerical values are synthetic placeholders.

Decoding setting	GSM8K			AIME25			GPQA Main		
	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE
Static k=1	66.4	1	0.029	12.4	1	0.045	20.4	1	0.043
Static k=2	61.4	2	0.043	11.8	2	0.055	19.6	2	0.052
Static k=3	53.9	3	0.057	11.1	3	0.065	17.9	3	0.061
Static k=4	51.0	4	0.071	10.4	4	0.075	16.1	4	0.07
Static k=5	42.9	5	0.085	9.4	5	0.085	14.4	5	0.079
Static k=6	38.2	6	0.099	8.4	6	0.095	12.6	6	0.088
Static k=7	33.9	7	0.113	7.5	7	0.105	10.9	7	0.097
Adaptive tau=0.995	64.8	1.96	0.037	12.7	2.91	0.052	20.7	1.5	0.048
Adaptive tau=0.99	64.2	2.17	0.038	12.3	3.12	0.054	20.3	1.5	0.049
Adaptive tau=0.98	63.5	2.37	0.04	11.9	3.32	0.055	20.0	1.67	0.051
Adaptive tau=0.97	62.7	2.58	0.041	11.5	3.53	0.057	19.6	1.88	0.052
Adaptive tau=0.96	62.3	2.68	0.043	11.1	3.63	0.059	19.3	1.98	0.054
Adaptive tau=0.95	61.4	2.89	0.044	10.6	3.84	0.061	18.9	2.19	0.055
Adaptive tau=0.90	58.8	3.39	0.045	10.2	4.34	0.062	18.6	2.69	0.056
Adaptive tau=0.87	57.1	3.69	0.047	9.8	4.64	0.064	18.2	2.99	0.058
Adaptive tau=0.85	56.4	3.8	0.048	9.4	4.75	0.066	17.9	3.1	0.059
Adaptive tau=0.80	54.5	4.1	0.05	9.0	5.05	0.067	17.6	3.4	0.061
Adaptive tau=0.75	52.4	4.41	0.051	8.5	5.36	0.069	17.2	3.71	0.062
Adaptive tau=0.70	50.2	4.71	0.052	8.1	5.66	0.071	16.9	4.01	0.063
Adaptive tau=0.65	47.0	5.12	0.054	7.7	6.07	0.072	16.5	4.42	0.065
Adaptive tau=0.60	44.5	5.42	0.055	7.3	6.37	0.074	16.1	4.72	0.066

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Table 9: Complete static and confidence-adaptive sweep for MTP with conditional full drift (M2). Accuracy/score, k_{eff} , and ECE are reported together for the three primary benchmarks. All numerical values are synthetic placeholders.

Decoding setting	GSM8K			AIME25			GPQA Main		
	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE
Static k=1	66.8	1	0.026	12.8	1	0.042	20.8	1	0.04
Static k=2	62.1	2	0.04	12.7	2	0.052	20.4	2	0.049
Static k=3	55.1	3	0.054	12.3	3	0.062	19.1	3	0.058
Static k=4	52.5	4	0.068	12.0	4	0.072	17.8	4	0.067
Static k=5	44.9	5	0.082	11.4	5	0.082	16.4	5	0.076
Static k=6	40.5	6	0.096	10.9	6	0.092	15.1	6	0.085
Static k=7	36.7	7	0.11	10.3	7	0.102	13.7	7	0.094
Adaptive tau=0.995	65.1	2.09	0.034	12.8	3.04	0.049	20.8	1.5	0.045
Adaptive tau=0.99	64.4	2.31	0.035	12.4	3.26	0.051	20.4	1.61	0.047
Adaptive tau=0.98	63.7	2.52	0.037	12.0	3.47	0.052	20.1	1.82	0.048
Adaptive tau=0.97	62.8	2.73	0.038	11.5	3.68	0.054	19.8	2.03	0.05
Adaptive tau=0.96	62.4	2.85	0.04	11.1	3.8	0.056	19.4	2.15	0.051
Adaptive tau=0.95	61.4	3.06	0.041	10.7	4.01	0.058	19.1	2.36	0.052
Adaptive tau=0.90	58.8	3.57	0.042	10.3	4.52	0.059	18.7	2.87	0.054
Adaptive tau=0.87	56.9	3.88	0.044	9.9	4.83	0.061	18.4	3.18	0.055
Adaptive tau=0.85	56.2	4.0	0.045	9.4	4.95	0.063	18.0	3.3	0.057
Adaptive tau=0.80	54.2	4.31	0.047	9.0	5.26	0.064	17.7	3.61	0.058
Adaptive tau=0.75	52.0	4.62	0.048	8.6	5.57	0.066	17.3	3.92	0.059
Adaptive tau=0.70	49.7	4.93	0.049	8.2	5.88	0.068	17.0	4.23	0.061
Adaptive tau=0.65	46.5	5.35	0.051	7.8	6.3	0.069	16.6	4.65	0.062
Adaptive tau=0.60	43.8	5.66	0.052	7.3	6.61	0.071	16.2	4.96	0.064

Table 10: Complete static and confidence-adaptive sweep for MTP with attraction-only drift (M3). Accuracy/score, k_{eff} , and ECE are reported together for the three primary benchmarks. All numerical values are synthetic placeholders.

Decoding setting	GSM8K			AIME25			GPQA Main		
	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE
Static k=1	66.6	1	0.028	12.6	1	0.044	20.6	1	0.042
Static k=2	61.7	2	0.042	12.2	2	0.054	20.1	2	0.051
Static k=3	54.5	3	0.056	11.7	3	0.064	18.5	3	0.06
Static k=4	51.7	4	0.07	11.2	4	0.074	17.0	4	0.069
Static k=5	43.9	5	0.084	10.4	5	0.084	15.4	5	0.078
Static k=6	39.3	6	0.098	9.6	6	0.094	13.9	6	0.087
Static k=7	35.3	7	0.112	8.9	7	0.104	12.3	7	0.096
Adaptive tau=0.995	64.9	2.04	0.031	12.8	2.99	0.046	20.7	1.5	0.043
Adaptive tau=0.99	64.3	2.25	0.032	12.4	3.2	0.048	20.4	1.55	0.044
Adaptive tau=0.98	63.5	2.46	0.034	11.9	3.41	0.049	20.0	1.76	0.046
Adaptive tau=0.97	62.7	2.67	0.035	11.5	3.62	0.051	19.7	1.97	0.047
Adaptive tau=0.96	62.2	2.78	0.037	11.1	3.73	0.053	19.3	2.08	0.049
Adaptive tau=0.95	61.3	2.99	0.038	10.7	3.94	0.054	19.0	2.29	0.05
Adaptive tau=0.90	58.7	3.5	0.039	10.3	4.45	0.056	18.6	2.8	0.051
Adaptive tau=0.87	56.9	3.81	0.041	9.8	4.76	0.058	18.3	3.11	0.053
Adaptive tau=0.85	56.2	3.92	0.042	9.4	4.87	0.06	17.9	3.22	0.054
Adaptive tau=0.80	54.2	4.23	0.044	9.0	5.18	0.061	17.6	3.53	0.056
Adaptive tau=0.75	52.1	4.54	0.045	8.6	5.49	0.063	17.2	3.84	0.057
Adaptive tau=0.70	49.8	4.85	0.046	8.2	5.8	0.065	16.9	4.15	0.058
Adaptive tau=0.65	46.5	5.25	0.048	7.7	6.2	0.066	16.5	4.55	0.06
Adaptive tau=0.60	44.0	5.56	0.049	7.3	6.51	0.068	16.2	4.86	0.061

Table 11: Complete static and confidence-adaptive sweep for MTP with block-level full drift (M4). Accuracy/score, k_{eff} , and ECE are reported together for the three primary benchmarks. All numerical values are synthetic placeholders.

Decoding setting	GSM8K			AIME25			GPQA Main		
	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE
Static k=1	66.9	1	0.025	12.9	1	0.041	20.9	1	0.039
Static k=2	62.4	2	0.039	12.8	2	0.051	20.6	2	0.048
Static k=3	55.4	3	0.053	12.6	3	0.061	19.4	3	0.057
Static k=4	53.0	4	0.067	12.4	4	0.071	18.1	4	0.066
Static k=5	45.4	5	0.081	11.9	5	0.081	16.9	5	0.075
Static k=6	41.2	6	0.095	11.4	6	0.091	15.6	6	0.084
Static k=7	37.4	7	0.109	11.0	7	0.101	14.4	7	0.093
Adaptive tau=0.995	65.1	2.13	0.028	12.8	3.08	0.043	20.8	1.5	0.041
Adaptive tau=0.99	64.5	2.35	0.029	12.4	3.3	0.045	20.5	1.65	0.042
Adaptive tau=0.98	63.7	2.56	0.031	12.0	3.51	0.046	20.1	1.86	0.043
Adaptive tau=0.97	62.8	2.78	0.032	11.6	3.73	0.048	19.8	2.08	0.045
Adaptive tau=0.96	62.3	2.89	0.034	11.1	3.84	0.05	19.4	2.19	0.046
Adaptive tau=0.95	61.4	3.11	0.035	10.7	4.06	0.052	19.1	2.41	0.048
Adaptive tau=0.90	58.7	3.63	0.036	10.3	4.58	0.053	18.7	2.93	0.049
Adaptive tau=0.87	56.9	3.94	0.038	9.9	4.89	0.055	18.4	3.24	0.05
Adaptive tau=0.85	56.1	4.06	0.039	9.5	5.01	0.057	18.0	3.36	0.052
Adaptive tau=0.80	54.1	4.37	0.041	9.0	5.32	0.058	17.7	3.67	0.053
Adaptive tau=0.75	51.9	4.69	0.042	8.6	5.64	0.06	17.3	3.99	0.054
Adaptive tau=0.70	49.5	5.0	0.043	8.2	5.95	0.062	17.0	4.3	0.056
Adaptive tau=0.65	46.2	5.42	0.045	7.8	6.37	0.063	16.6	4.72	0.057
Adaptive tau=0.60	43.6	5.73	0.046	7.4	6.68	0.065	16.3	5.03	0.059

Table 12: Complete static and confidence-adaptive sweep for Soft on-policy KL (M5). Accuracy/score, k_{eff} , and ECE are reported together for the three primary benchmarks. All numerical values are synthetic placeholders.

Decoding setting	GSM8K			AIME25			GPQA Main		
	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE	Acc. (%)	k_{eff}	ECE
Static k=1	66.2	1	0.031	12.2	1	0.047	20.2	1	0.045
Static k=2	61.1	2	0.045	11.5	2	0.057	19.3	2	0.054
Static k=3	53.5	3	0.059	10.7	3	0.067	17.5	3	0.063
Static k=4	50.5	4	0.073	9.8	4	0.077	15.6	4	0.072
Static k=5	42.3	5	0.087	8.8	5	0.087	13.8	5	0.081
Static k=6	37.5	6	0.101	7.7	6	0.097	11.9	6	0.09
Static k=7	33.1	7	0.115	6.7	7	0.107	10.1	7	0.099
Adaptive tau=0.995	64.7	1.94	0.025	12.7	2.89	0.04	20.7	1.5	0.038
Adaptive tau=0.99	64.0	2.14	0.026	12.3	3.09	0.042	20.3	1.5	0.039
Adaptive tau=0.98	63.3	2.34	0.028	11.9	3.29	0.043	20.0	1.64	0.041
Adaptive tau=0.97	62.5	2.55	0.029	11.5	3.5	0.045	19.6	1.85	0.042
Adaptive tau=0.96	62.1	2.65	0.031	11.0	3.6	0.047	19.3	1.95	0.044
Adaptive tau=0.95	61.2	2.85	0.032	10.6	3.8	0.049	18.9	2.15	0.045
Adaptive tau=0.90	58.7	3.35	0.033	10.2	4.3	0.05	18.6	2.65	0.046
Adaptive tau=0.87	56.9	3.66	0.035	9.8	4.61	0.052	18.2	2.96	0.048
Adaptive tau=0.85	56.3	3.76	0.036	9.4	4.71	0.054	17.9	3.06	0.049
Adaptive tau=0.80	54.4	4.06	0.038	8.9	5.01	0.055	17.5	3.36	0.051
Adaptive tau=0.75	52.3	4.36	0.039	8.5	5.31	0.057	17.2	3.66	0.052
Adaptive tau=0.70	50.1	4.67	0.04	8.1	5.62	0.059	16.8	3.97	0.053
Adaptive tau=0.65	47.0	5.07	0.042	7.7	6.02	0.06	16.5	4.37	0.055
Adaptive tau=0.60	44.5	5.37	0.043	7.3	6.32	0.062	16.1	4.67	0.056

Table 13: Recommended validation-time latent alignment report. The statistics are computed between normalized teacher and student feature populations after the same PCA preprocessing used throughout validation. NN1 is a two-sample nearest neighbor accuracy and should approach 50%.

Metric	Primary interpretation	Desired trend
SW ²	Projected transport discrepancy	↓
MMD	Kernel two-sample discrepancy	↓
Sinkhorn	Entropically regularized OT cost	↓
NN1	1-NN two-sample classification accuracy	→ 50%